

Algebra II with Trigonometry

Chapter 11.1

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4652951_alg-2trig-h-chp-11-1-note-docx.pdf](#)

Problems

1. A circle centered at the origin is tangent to the parabola $y^2 = 8x$. Find the radius of the circle.
2. The parabola $x^2 = 4py$ and the circle $x^2 + (y - 3)^2 = 9$ intersect in exactly one point. Find p .
3. A point P lies on the parabola $y^2 = 12x$. The distance from P to the focus equals 10. Find the coordinates of P .
4. Two parabolas share the same vertex at the origin: $x^2 = 4ay$ and $y^2 = 4ax$, where $a > 0$. Find the area enclosed by their arcs.
5. A circle centered at $(2, 0)$ passes through the focus of the parabola $y^2 = 8x$ and is tangent to its directrix. Find the equation of the circle.
6. Let P be a point on the parabola $x^2 = 4y$. A line through P with slope 1 meets the directrix at A and the y -axis at B . Find the value of AB , independent of the choice of P .